

The suprarenal (adrenal) glands lie immediately superior and slightly anterior to the upper pole of each kidney (see Figs 74.5, 62.8). Golden yellow in colour, each gland possesses two functionally and structurally distinct areas: an outer cortex and an inner medulla. The glands are surrounded by perinephric fat enclosed within the renal fascia, and separated from the kidneys by a small amount of fibrous tissue.

The dimensions of the suprarenal glands in adults *in vivo* have been measured from axial computed tomography (CT) scans (Vincent et al 1994). The mean maximum width of the body of the suprarenal gland is 61 mm (right) and 79 mm (left), and the mean width of each limb of the gland (medial and lateral) is approximately 30 mm. No individual suprarenal limb should measure more than 5 mm in transverse section. In adults, each suprarenal gland weighs approximately 5 g (the medulla contributes about one-tenth of the total weight) and has a volume of approximately 3–6 cm<sup>3</sup> (Wang et al 2013).

The glands are macroscopically slightly different in external appearance (Fig. 71.1). The right is pyramidal in shape and has two well-developed lower projections (limbs), giving a cross-sectional appearance similar to a three-pointed star. The left gland is more semilunar in

shape, flattened in the anteroposterior plane and marginally larger than the right. The bulk of the right suprarenal sits on the apex of the right kidney and usually lies slightly higher than the left gland, which lies on the anteromedial aspect of the upper pole of the left kidney.

At birth, the glands are proportionately larger and are approximately one-third the size of the ipsilateral kidney. The cortex of each gland is expanded by a well-developed 'fetal zone', responsible for producing dehydroepiandrosterone (DHEA), the substrate for placental oestrogen, in the fetus. The suprarenal cortex (specifically the fetal zone) reduces in size immediately after birth and the medulla grows comparatively little. By the end of the second postnatal month, the weight of the suprarenal has reduced by more than 50%. The glands begin to grow again by the end of the second year and regain their weight at birth by puberty. There is little further weight increase in adult life.

Small accessory suprarenal nodules, composed mainly of cortical tissue (also known as 'adrenal rests'), may occur in the areolar tissue near the suprarenal glands. They are also occasionally found in the spermatic cord, epididymis or testis in boys (Altın et al 1992), and ovary or broad ligament of the uterus in girls. Ectopic suprarenal tissue may cause diagnostic confusion and is rarely the site of neoplastic change.

## RIGHT SUPRARENAL GLAND

The right suprarenal gland lies posterior to the inferior vena cava, separated from it by only a thin layer of fascia and connective tissue. It also lies posterior to the right lobe of the liver and anterior to the right crus of the diaphragm and superior pole of the right kidney (Fig. 71.2). Its inferior surface or base overlaps the anterosuperior aspect of the upper pole of the right kidney, with the two lower projections (limbs) of the gland straddling the renal tissue. The anterior surface has two distinct facets: a narrow medial facet that lies posterior to the inferior vena cava, and a triangular lateral facet that lies in contact with the bare area of the liver. The lowest part of the anterior surface may be covered by the peritoneal reflection of the inferior layer of the coronary ligament, which also represents the upper recess of the hepatorenal pouch (see Fig. 63.2). Here, the gland lies posterior to the lateral border of the second part of the duodenum. Below the apex, near the anterior border of the gland, the right suprarenal vein emerges from the hilum to join the inferior vena cava. This vein is short, which makes surgical resection of the gland or mobilization of the inferior vena cava potentially hazardous. It may be inadvertently avulsed from the inferior vena cava during surgery or, occasionally, by high-energy deceleration injuries. The posterior surface of the gland is divided into upper and lower areas by a curved transverse ridge. The large upper area is slightly convex and abuts the diaphragm, whereas the small lower area is concave and lies in contact with the superior aspect of the upper pole of the right kidney. The medial border of the gland is thin and lies lateral to the right coeliac ganglion and the right inferior phrenic artery where the artery runs over the right crus of the diaphragm.

## LEFT SUPRARENAL GLAND

The left suprarenal gland is closely applied to the left crus of the diaphragm, separated from it by a thin layer of fascia (see Fig. 71.2). The medial aspect is convex longitudinally, whereas the lateral aspect is concave because it is moulded by the medial surface of the upper pole of the left kidney. The superior border is sharply defined while the inferior surface is more rounded. The upper part of the anterior surface is covered by the peritoneum of the posterior wall of the lesser sac, which separates it from the stomach and sometimes from the postero-inferior border of the spleen. A small lower part of the anterior surface

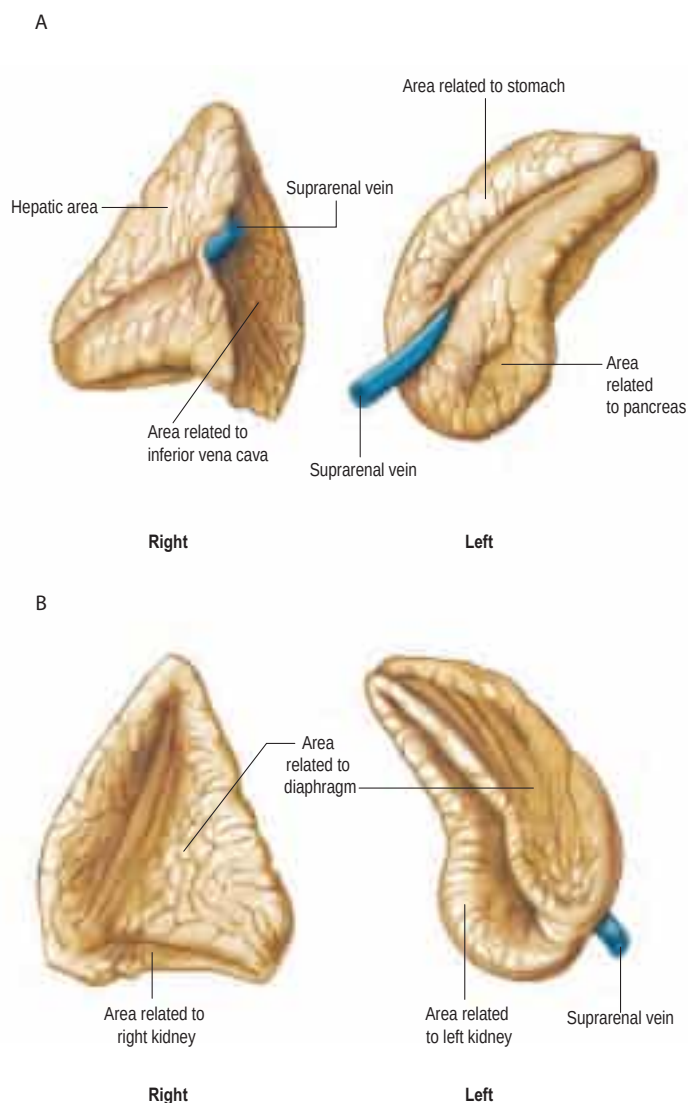
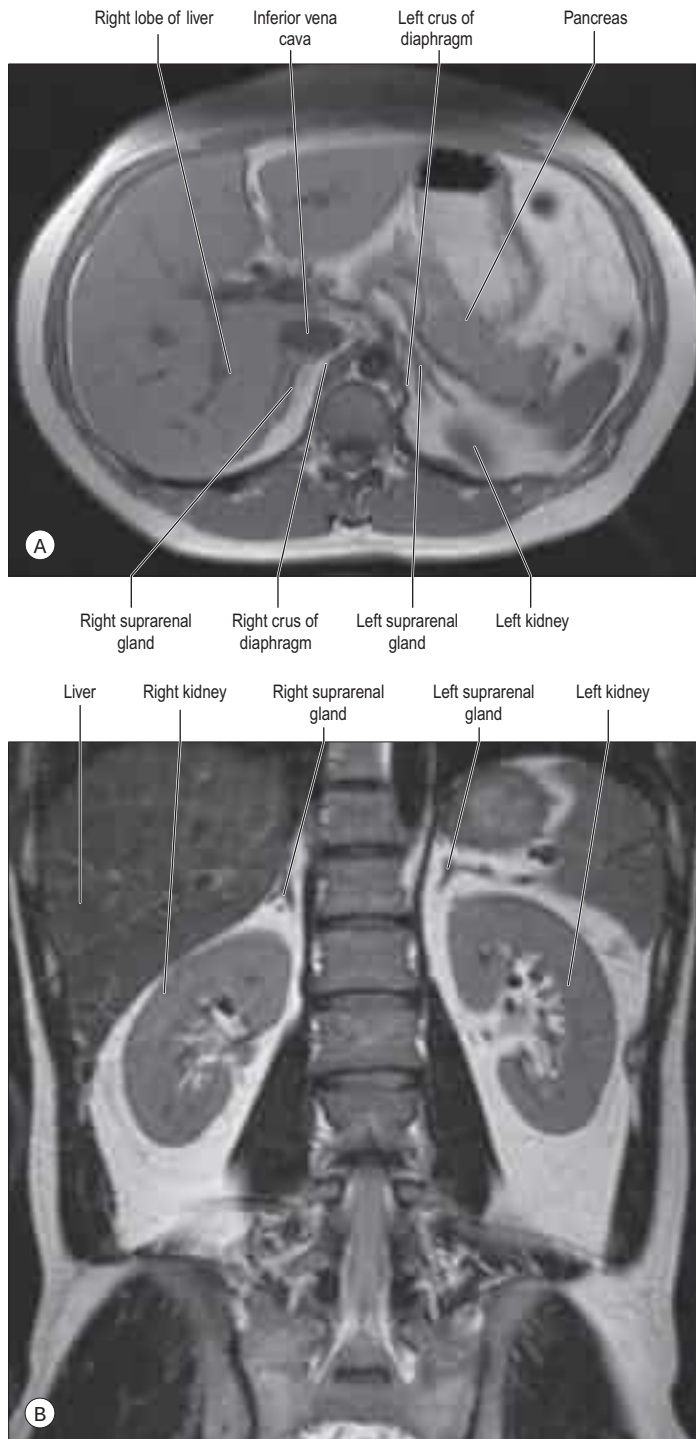


Fig. 71.1 The suprarenal glands. A, Anterior aspect. B, Posterior aspect.



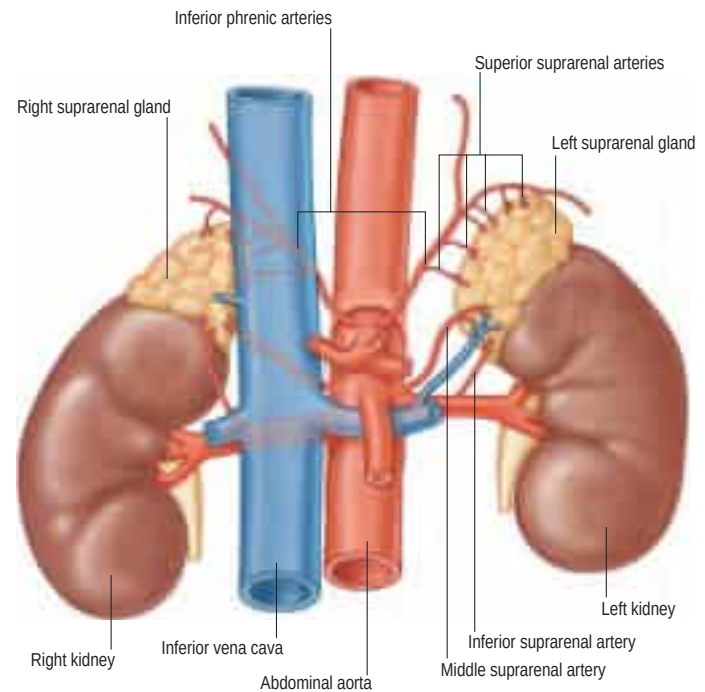
**Fig. 71.2** Magnetic resonance imaging (MRI) of the suprarenal glands. A, Axial T1 weighted MRI. B, Coronal T2 weighted MRI. (Courtesy of Dr Louise Moore, Chelsea & Westminster Hospital.)

is not covered by peritoneum and lies adjacent to the pancreas and splenic artery. The hilum faces inferiorly from the lower part of the medial aspect. The left suprarenal vein emerges from the hilum and runs inferomedially to join the left renal vein. The posterior surface of the gland is divided by a ridge into a lateral area adjoining the kidney and a smaller medial area that lies in contact with the left crus of the diaphragm. The medial border lies lateral to the left coeliac ganglion and the left inferior phrenic and left gastric arteries, which ascend on the left crus of the diaphragm.

## VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

### Arteries

The suprarenal glands have one of the highest arterial flow rates per gram of tissue (up to 5 ml/g/min) (Williams and Leggett 1989). Each gland is supplied by superior, middle and inferior suprarenal arteries, whose main branches may be multiple (Fig. 71.3). They ramify over



**Fig. 71.3** The arterial supply and venous drainage of the suprarenal glands. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

the capsule before penetrating the gland to form a subcapsular arterial plexus, from which fenestrated sinusoids pass around clustered glomerulosa cells and between the columns in the zona fasciculata to a deep plexus in the zona reticularis. Venules from this plexus pass between medullary chromaffin cells to medullary veins, which they enter between prominent bundles of smooth muscle fibres. Some relatively large arteries bypass this indirect route and pass directly to the medulla (Fig. 71.4). The arterial supply of the suprarenal gland is prone to considerable variation (Manso and DiDio 2000, Dutta 2010); only the main variants will be described here.

### Superior suprarenal arteries

The superior suprarenal artery usually arises from the ipsilateral inferior phrenic artery and passes to the gland as four or five small branches; it may occasionally arise from the abdominal aorta.

### Middle suprarenal arteries

The middle suprarenal artery is usually single, but may be multiple or absent. It often arises from the lateral aspect of the abdominal aorta at around the level of the superior mesenteric artery and ascends slightly over the crura of the diaphragm to anastomose with the other suprarenal arteries on the surface of the gland. The right middle suprarenal artery passes behind the inferior vena cava close to the right coeliac ganglion, whereas the left middle suprarenal artery passes close to the left coeliac ganglion, splenic artery and the superior border of the pancreas. The middle suprarenal artery may originate from either the ipsilateral inferior phrenic or renal artery.

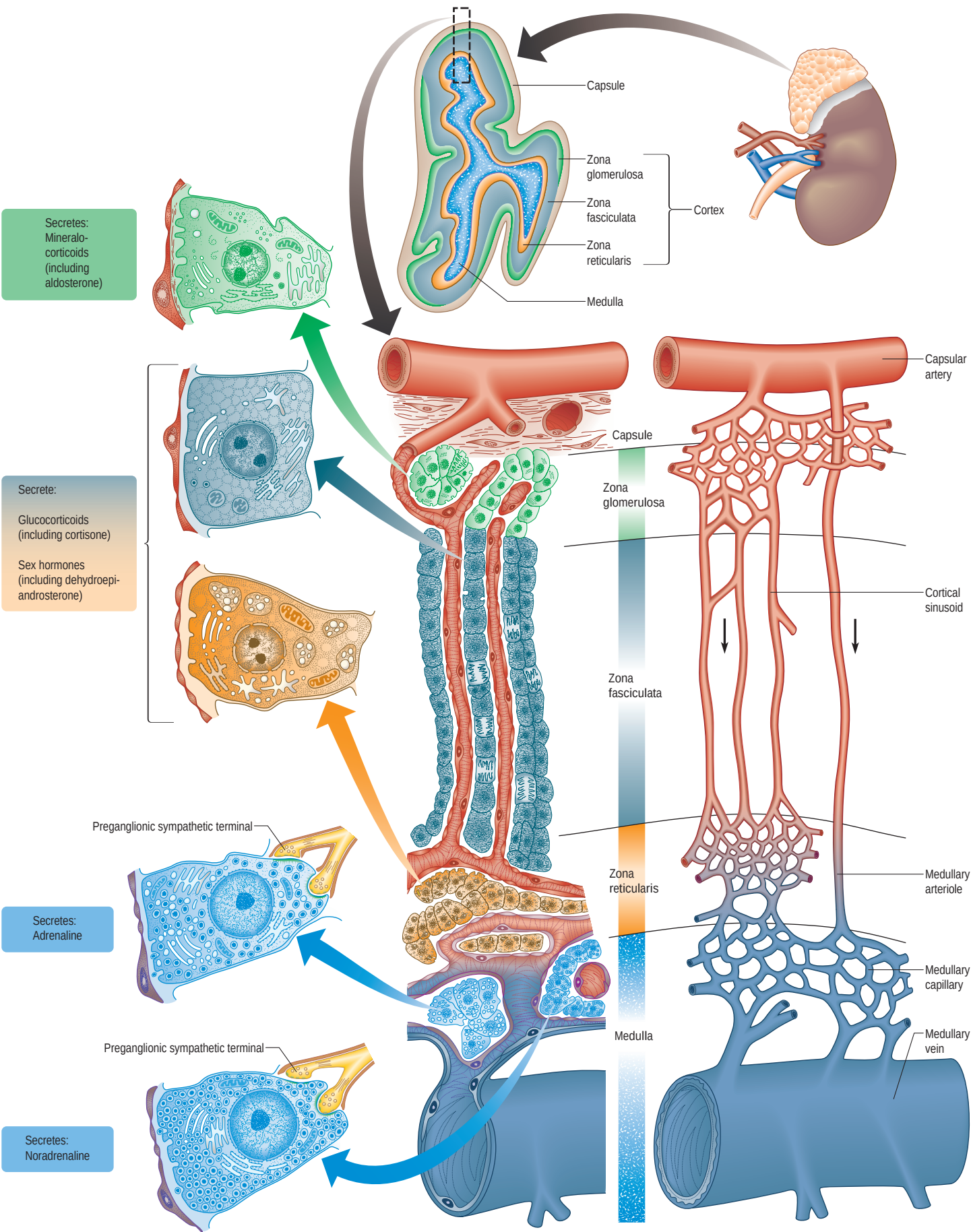
### Inferior suprarenal arteries

The inferior suprarenal arteries often contribute most of the arterial supply to the gland. One or two arteries usually arise from the ipsilateral renal artery, but the inferior suprarenal arteries may originate from either the abdominal aorta or the ipsilateral gonadal artery.

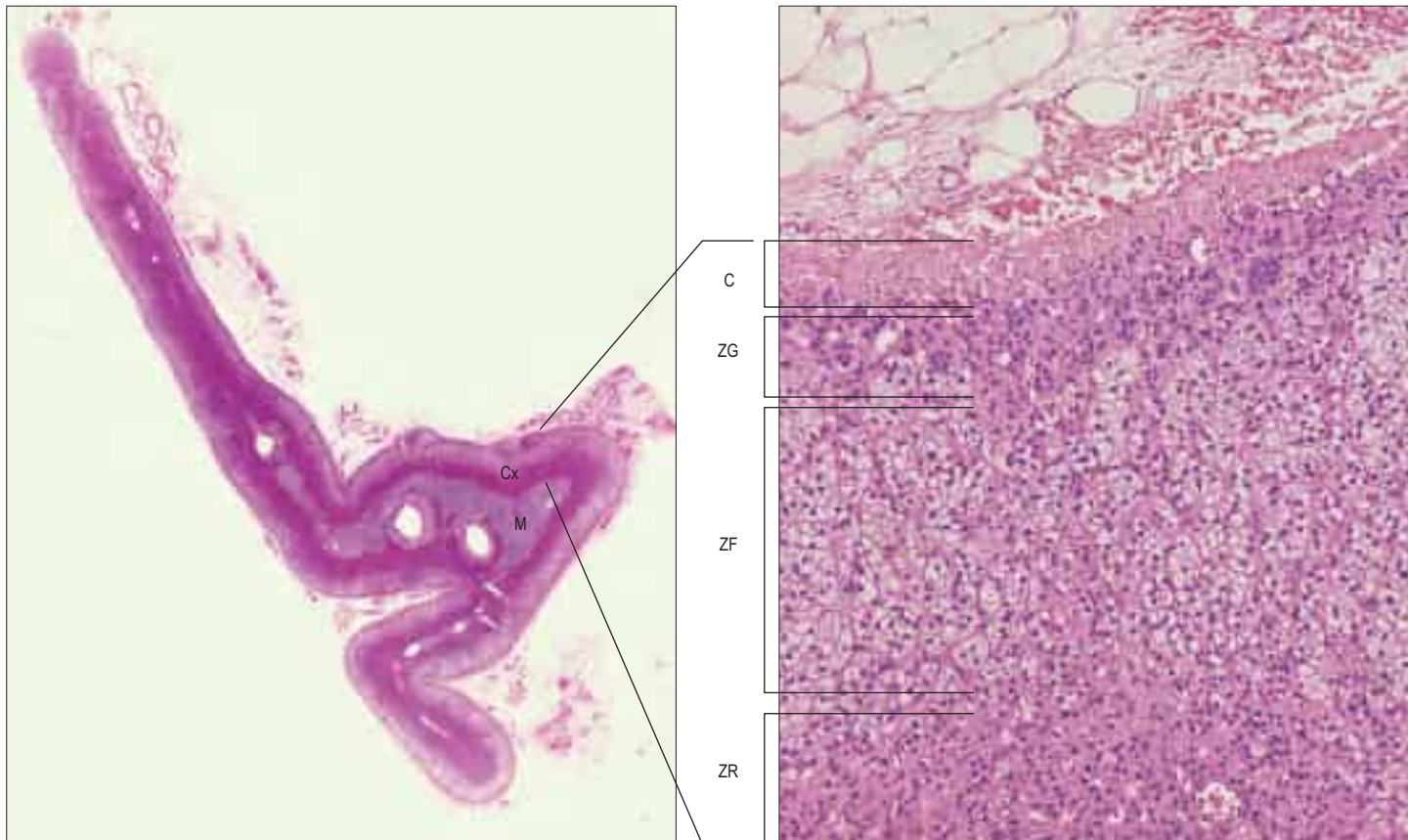
### Veins

Medullary veins emerge from the hilum to form a suprarenal vein, which is usually single. The right vein is very short, and passes directly and horizontally into the posterior aspect of the inferior vena cava. An accessory right suprarenal vein is occasionally present and runs from the hilum superomedially to join the inferior vena cava above the right suprarenal vein. The short course renders the right suprarenal vein(s) vulnerable to injury or even avulsion from the inferior vena cava during surgery if undue traction is applied. The left suprarenal vein is longer and descends medially, anterolateral to the left coeliac ganglion, then passes posterior to the pancreatic body and usually drains into the left renal vein; it may receive the left inferior phrenic vein (Loukas et al 2005). Since the venous drainage from each gland is usually via a single





**Fig. 71.4** The gross sectional appearance, microstructure, vasculature and ultrastructure of the suprarenal gland. Brief functional summaries are included.



**Fig. 71.5** A cross-section through an adult suprarenal gland showing the cortex (Cx) and highly vascular medulla (M). The cortex is shown at higher magnification to demonstrate the outer capsule (C), the zona glomerulosa (ZG), zona fasciculata (ZF) and zona reticularis (ZR). The boundaries between zones are indistinct and variable. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

vein, infarction of the gland is more likely to be caused by damage to a suprarenal vein than to one of the suprarenal arteries.

Understanding variant suprarenal venous anatomy is important to avoid bleeding during minimally invasive adrenalectomy, particularly in patients with large tumours and pheochromocytomas. Cadaver studies have shown little variation in suprarenal venous anatomy but suggest that variants are more commonly found on the right (Cesmebasi et al 2014). However, a small clinical series showed that there may be a higher rate of suprarenal vein variants in patients with pheochromocytoma (Parnaby et al 2008).

Anticipating variant suprarenal venous anatomy is key to minimizing bleeding during laparoscopic adrenalectomy, particularly in patients with large tumours or pheochromocytomas. Sholten et al (2013) reported variant suprarenal veins in 13% of individuals undergoing laparoscopic adrenalectomy, more often on the right (17%) than the left (9%). Common variants included an absent main suprarenal vein; two or more suprarenal veins draining one side; and veins draining to the inferior phrenic vein or, on the left, to the inferior vena cava.

### Lymphatic drainage

Lymphatic drainage within the capsule of the suprarenal gland communicate with subserous lymphatics that drain medially to para-aortic and paracaval nodes (Merklin 1966). A few capsular lymphatics communicate with lymph vessels that pass through the diaphragm.

### INNERVATION

The suprarenal gland, relative to its size, has a greater autonomic supply than any other organ. The nerves are distributed throughout the gland: around blood vessels (regulating blood flow), in the medulla (stimulating the release of catecholamines from chromaffin cells), and in the cortex (where they may influence steroid hormone production; Tóth et al (1997)). A suprarenal plexus lies between the medial aspect of each gland and the coeliac and aorticorenal ganglia. It contains predominantly preganglionic sympathetic fibres that originate in the lower thoracic spinal segments, reach the plexus via branches of the greater splanchnic nerves, and synapse on clusters of large medullary chromaffin cells (which can be regarded as homologous with postganglionic

sympathetic neurones). A smaller proportion of postganglionic sympathetic nerve fibres innervate cortical blood vessels. Other nerve fibres containing a variety of neurotransmitters have been identified within the suprarenal cortex and may be involved in modulating steroid hormone secretion. The cell bodies of afferent nerve fibres arising in the suprarenal medulla are almost all located in dorsal root ganglia, although some lie in vagal ganglia; their precise function is unknown (Mravec 2005).

### MICROSTRUCTURE

In section, the suprarenal gland has an outer cortex, which is yellowish in colour and forms the main mass, and a thin medulla, forming about one-tenth of the gland, which is dark red or greyish, depending on its content of blood (see Fig. 71.4; Fig. 71.5). The medulla is completely enclosed by cortex, except at the hilum. The gland has a thick collagenous capsule from which trabeculae extend deep into the cortex. The capsule contains a rich arterial plexus (see above) that supplies branches to the gland.

### Suprarenal cortex

The suprarenal cortex is composed of a zona glomerulosa, a zona fasciculata and a zona reticularis (see Fig. 71.5). The outer, subcapsular, zona glomerulosa consists of a narrow region of small polyhedral cells arranged in rounded clusters. The cells have deeply staining nuclei and a basophilic cytoplasm containing a few lipid droplets. Ultrastructurally, their cytoplasm displays an abundant smooth endoplasmic reticulum that is typical of steroid-synthesizing cells. The broader, intermediate, zona fasciculata consists of large polyhedral basophilic cells arranged in straight columns, two cells wide, separated by parallel fenestrated venous sinusoids. The cells contain many lipid droplets and large amounts of smooth endoplasmic reticulum. The innermost part of the cortex, the zona reticularis, consists of branching interconnected columns of rounded cells with cytoplasm containing smooth endoplasmic reticulum, numerous lysosomes and aggregates of brown lipofuscin pigment that accumulate with age.



Cortical cells produce several hormones, and the cells of the zonae fasciculata and reticularis are also rich in ascorbic acid. Cells in the zona glomerulosa produce mineralocorticoids, e.g. aldosterone, which regulates electrolyte and water balance; cells in the zona fasciculata produce hormones maintaining carbohydrate balance (glucocorticoids), e.g. cortisol (hydrocortisone); and cells in the zona reticularis produce sex hormones (progesterone, oestrogens and androgens). The suprarenal cortex is essential to life and its complete removal is lethal without replacement therapy. It exerts considerable control over lymphocytes and lymphoid tissue; an increased secretion of corticosteroids can result in a marked reduction in lymphocyte numbers.

The deeper part of the zona fasciculata widens in pregnancy. Cortical atrophy in elderly males is greatest in the same region.

### Suprarenal medulla

The suprarenal medulla is composed of groups and columns of chromaffin cells (phaeochromocytes), separated by wide venous sinusoids and supported by a network of reticular fibres. Chromaffin cells, so called from their colour reaction to dichromate fixatives, form part of the neuroendocrine system and are functionally equivalent to postganglionic sympathetic neurones. They synthesize, store (as granules) and release the catecholamines noradrenaline (norepinephrine) and adrenaline (epinephrine) into the venous sinusoids. Release is under preganglionic sympathetic control, mediated by the sympathetic neurones that occur either singly or in small groups in the medulla.

The majority of chromaffin cells synthesize adrenaline and store it in small granules with a dense core. Less numerous noradrenaline-secreting cells have larger granules with a dense eccentric core. Some cells synthesize both hormones. Chromogranin proteins package catecholamines within the granules, which also contain enkephalins, opiate-like proteins that may have endogenous analgesic effects in some circumstances. All of the cells are large, with large nuclei and basophilic, faintly granular cytoplasm. They form single rows along the venous sinusoids. Sympathetic axon terminals synapse with the chromaffin cells on the surfaces that face away from the sinusoids.

The sinusoids are lined by fenestrated endothelium and drain to the central medullary vein and hilar suprarenal vein. Under normal circumstances, little adrenaline or noradrenaline is released; secretion is increased in response to fear, anger or stress. Unlike the cortex, the suprarenal medulla is not essential to life.

### SUPRARENAL GLAND EXCISION

Removal of one or both of the suprarenal glands may be performed using 'open surgery', which remains the standard of care for patients with adrenocortical carcinoma who require *en bloc* resection of the

tumour. For other tumours and disorders of the suprarenal gland requiring surgery, minimally invasive techniques are now generally preferred. The original laparoscopic approaches were described in 1992 (transabdominal; [Gagner et al 1992](#)) and 1995 (posterior retroperitoneal; [Merican et al 1995](#)). The posterior approach is considered superior on account of its safety, rapidity, avoidance of the peritoneal cavity, and the ability to operate on both sides without repositioning the patient ([Morris and Perrier 2012](#)). This has been confirmed in several large cohort studies ([Walz et al 2006](#), [Dickson et al 2011](#)).

Nevertheless, conventional laparoscopy has certain disadvantages, such as a two-dimensional view, unstable camera platform, and rigid instrumentation. Robotic surgery is a new and emerging technique now performed in many centres and offers the advantages of a three-dimensional stable platform, seven degrees of freedom, and enhanced vision ([Taskin and Berber 2013](#)). During the last decade, experience with robotic adrenalectomy has highlighted several facets, including a steep learning curve, no significant reduction in the need to convert to an open procedure, operative complications or blood loss compared to conventional laparoscopy, and longer operative times (although this decreases with experience). Robotic surgery is also more expensive. However, with appropriate patient selection, robotic adrenalectomy can be advantageous. For example, patients with familial suprarenal disorders, who are more likely to have bilateral pathology, require cortical sparing, or have glands that may be difficult to access posteriorly, are particularly good candidates for robotic surgery.

In the posterior retroperitoneal approach, the patient is placed in the prone jack-knife position to open up the interval between the costal margin and iliac crest posteriorly. A small incision is made 2 cm inferior and parallel to the twelfth rib, and the perinephric fascia is entered using blunt dissection or with the aid of laparoscopic visualization. A dissecting balloon is then inserted and inflated under laparoscopic control, after which a 12 mm trocar is inserted into the cavity and CO<sub>2</sub> is insufflated up to a pressure of 20–25 mmHg. The 0° laparoscope is replaced by a 45° telescope, and two additional 5- or 10 mm ports are inserted medial and lateral to the initial port. Dissection is initiated at the superior aspect of the suprarenal gland and proceeds laterally and then inferiorly. The medial surface of the gland is dissected last and the adrenal vessels isolated and divided either with clips or a harmonic scalpel ([Lal and Clark 2010](#)).

Similar steps are taken in the robotic approach. The robot is docked after insertion of the 5- and 10 mm secondary ports; the robotic grasper is placed in the lateral port and the harmonic scalpel in the medial before proceeding with the dissection.

In obese (or Cushingoid) patients, identification of a relatively normal-sized suprarenal gland can be extremely difficult and time-consuming. The suprarenal vein emerges from the lower medial border of the gland and is often a very substantial structure. In contrast, the supplying arteries tend to be small and named arteries are often difficult to identify during surgery.

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